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Patterns of intrinsic capacity among community-dwelling older adults: Identification by latent class analysis and association with one-year adverse outcomes

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ABSTRACT

Objective: To identify patterns of intrinsic capacity (IC) and determine the association between these patterns with incident one-year outcomes.

Methods: A total of 756 older adults aged ≥ 60 years were followed up after 1 year. IC was assessed using the revised integrated care for older people screening tool, and its patterns were examined by the latent class analysis. Logistic regression models were conducted to compare the risk of adverse outcomes.

Results: Three IC patterns were identified. Both “sharp declines in sensory domain” (Class 2) and “sharp declines in locomotion, psychological, cognition and vitality domains” (Class 3) were at greater risk of disabilities and poor physical quality of life than “relatively healthy” (Class 1). The Class 3 was twice as likely to be hospitalized as Class 1.

Discussion: Assessment of IC could provide valuable information on stratifying older adults into heterogeneous groups, promoting targeted interventions to delay the adverse outcomes.

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Introduction

Population aging and geriatric health promotion have emerged as global challenges. As people age, their health problems tend to become more chronic and complex. The existing disconnected and fragmented healthcare, with a singular focus on specific conditions, fails to satisfy their demands, resulting in enormous costs and burden for older adults and healthcare system.¹ Therefore, the World Health Organization (WHO) published *Integrated care for older people: Guidelines on community-level interventions to manage*, which formulates a person-centered approach with the ultimate aim of maximizing individuals' intrinsic capacity and promoting healthy ageing. Derived from the *World report on ageing and health*, intrinsic capacity is defined as “the composite of all the physical and mental capacities of an individual”.² It is a multidimensional structure containing five domains (i.e., cognition, locomotion, vitality, sensory function, and psychological domain), where these domains interact with each other and the specific combination of impaired domains may have more impact on physical function than the sum of individually

impaired domains.³ As sensitive dynamic indicators of healthy ageing, the declines in intrinsic capacity occur much earlier before the emergence of age-associated diseases or symptoms.⁴ Thus, as the target of health management for older people, they may serve as excellent predictors of adverse outcomes to identify older adults at high risk and delay the onset of care dependence through effective measures.

However, among previous research on the predictive value of intrinsic capacity on adverse outcomes, one part took the total score as predictors,^{5,6} hindering the implementation of further targeted intervention; the other part took specific intrinsic capacity domains as predictors, ignoring the interaction between the impaired domains and the population heterogeneity caused by the interaction of multiple factors.^{6,7} Briefly, each separated domain or simply calculating the total score of all intrinsic capacity domains fails to comprehensively reflect the complex health status of heterogeneous population and efficiently guide the next intervention. However, the intrinsic capacity patterns of older adults identified by the latent class analysis (LCA) may shed light on this issue. The LCA is a statistical approach that permits model-based clustering of heterogeneous populations by specific combinations of observed variables.⁸ Relative to traditional statistical methods, the LCA has the statistical power for investigating co-exposure, and excavates combined influences masked by collinearity of independent variables.⁹ Through the LCA, the best

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latent classes of intrinsic capacity in older adults can be identified based on the declines in different domains. From the perspective of population-level interventions, it is conducive to implementing classified management for older adults and maximizing the effectiveness of intervention under the limited medical development level and available resources. At the level of individual interventions, classifying individuals by intrinsic capacity patterns to small heterogeneous subgroups allows for tailoring targeted interventions and fulfilling specific needs.

To our knowledge, there is no research on identifying the latent classes of intrinsic capacity through the LCA and exploring the predictive value of the latent classes on adverse outcomes. The LCA has been used to identify multimorbidity patterns including healthy group and various multimorbidity groups with specific diseases that were associated with adverse outcomes, such as disability, quality of life (QoL), mortality and healthcare utilization.^{10,11} Nevertheless, instead of focusing on the presence of individual diseases or even the extent of comorbidities, WHO advocates comprehensive functional assessments of older adults to better grasp and meet their health needs, which calls attention to the cluster analysis on functions.² A previous study based on a biopsychosocial framework conducted LCA on indicators from multiple domains, including sensory function, motor function, activities of daily living, health status, mental functioning, personal factors and environmental factors, and identified seven heterogeneous functioning profiles among participants aged 60 years or older.¹² Although the multidimensional nature of functions was considered in this study, the multifarious and complex indicators limit its generalization or comparison with other research results. Furthermore, WHO has proposed comprehensive assessments and integrated care for older adults (ICOPE) in extensive community health practices.¹³ Thus, performing LCA in the indicators of the ICOPE screening tool within the framework of intrinsic capacity to identify the functional patterns may have a better prospect of widespread application in research and practice.

Therefore, the aims of this study were to identify the patterns of intrinsic capacity among community-dwelling older adults using LCA, and to compare the risk of one-year adverse outcomes, including disability, hospitalization and poor QoL, among different classes. We hypothesized that different patterns of declines in intrinsic capacity would be identified among older adults, and subgroups with sharp declines in specific domains would have higher risks of one-year adverse outcomes, including disabilities, hospitalization and poor QoL.

Material and methods

Study design and setting

This was a prospective study with 1-year follow-up conducted among Chinese community-dwelling older adults in Jinan city of Shandong Province in Eastern China. The ethical approval was granted from the Institutional Review Board of the researchers' university (No. LL-201601029). All participants provided written informed consent.

Participants

Baseline data were collected from July to December in 2016 (Wave 1). Participants were enrolled using a multistage stratified sampling. First, according to gross domestic product of Jinan City in 2015, 3 of 9 districts within Jinan City were randomly selected to represent high (Lixia district), medium (Shizhong district), and low economic level (Tianqiao district), respectively. Next, a total of 22 communities (geographic areas as defined by the government) were included from these three districts on the basis of sample size of older

population and approval from the community residents' committees. Then, potential participants were invited to their affiliated community centers for further survey. The inclusion criteria were as follows: (a) aged 60 years and above; (b) living in these 22 selected communities; (c) volunteering to participate in the survey. The exclusion criteria were as follows: (a) suffering severe cognitive impairment screened by the Short Portable Mental Status Questionnaire (SPMSQ) with a score ≥ 8 ;¹⁴ (b) having serious physical deficits, such as in a wheelchair. Of the 1260 residents invited, 25 were excluded because they had serious physical deficits or severe cognitive impairment. Therefore, a total of 1235 participants underwent face-to-face structured questionnaire interviews and objective physical performance measurements.

After one year, follow-up data were collected from July to October in 2017 (Wave 2). Of the 1235 participants at baseline, 24 moved elsewhere, 281 were out of contact, 149 rejected further participation with unknown reasons, 6 died, and 12 withdrew due to poor health such as poor cardiac function and lumbar sprain. Finally, 781 participants' one-year adverse outcomes were available in Wave 2. Participant flow diagram has been published in our previously published paper.⁵³

Measures

Intrinsic capacity

The intrinsic capacity screening tool was adapted from the ICOPE screening tool¹³ on the basis of available data. Cognition domain was assessed by the SPMSQ, a 10-item questionnaire involving the assessment of memory and orientation recommended by ICOPE.¹⁴ Locomotion domain was evaluated by chair rise test as recommended by ICOPE.¹³ Vitality domain was assessed using two single-item questions concerning weight loss and appetite loss. The sensory domain was measured using self-report hearing and visual function. The psychological domain was evaluated by two items (i.e., "Have you felt down in the past week?", and "Have you felt that you could not get going?"). Each domain of intrinsic capacity was scored as 0 if intact and otherwise as 1. More details has been described in our previously published paper.⁵³

Adverse outcomes

Disability in activities of daily living (ADL) was assessed using the Katz Index. Incident disability in ADL was defined using the Edwards-Nunnally method which assesses clinically individual change and evades the problem of regression to the mean.¹⁵ Following standard practice, we included the reliability of the scale to calculate an adjusted score in Wave 1, and built a confidence interval around the adjusted score taking $p < 0.10$ as a criterion value for each participant.¹⁶ According to the established procedure, incident disability in ADL was recognized when participants' score on the Katz Index¹⁷ in wave 2 was below the lower confidence interval. Disability in instrumental activity of daily living (IADL) was assessed using the Lawton Scale.¹⁸ QoL was measured by the 12-item Short Form Health Survey (SF-12),¹⁹ including physical component summary (PCS) and mental component summary (MCS). Incident disability in IADL, poor PCS, and poor MCS were also defined using the aforementioned Edwards-Nunnally method. Hospitalization was defined if participants had been admitted to hospital for any causes (e.g., surgically or medically related conditions) during the past year.

Covariates

Sociodemographic characteristics were collected including age, sex, marital status, educational level, and monthly income at baseline. Multimorbidity was recognized as the concurrent presence of two or more chronic diseases. Social support was measured using the 10-item Social Support Rating Scale.²⁰

Statistical analysis

Data analysis was implemented in Mplus 8.4 and SPSS 24.0. The level of statistical significance was set at $p < 0.05$. According to previous research,²¹ 11 participants with poor health such as lumbar sprain and 6 who passed away were classified as having incident ADL and IADL disabilities, and 1 participant suffering cardiac dysfunction as experiencing hospitalization. Individual cases were deleted if there were one or more missing data in any intrinsic capacity domain and covariates.

The LCA was performed to examine patterns of declines in intrinsic capacity among community-dwelling older adults. A series of increasingly complex models (adding classes) were estimated to determine the optimal model. In terms of selection criteria, models are best fitted when Bayesian Information Criterion (BIC) and adjusted BIC (aBIC) scores are lower; a reduction of BIC or aBIC > 10 between models indicates the improvement of goodness-of-fit.²² Entropy value informs classification accuracy, with value ≥ 0.8 being considered excellent.²³ Average posterior probabilities belonging to the assigned class above 0.70 indicate optimal fitness. Apart from the above fit indices, substantive interpretability and parsimony should also be taken into consideration. Furthermore, each class should comprise at least 5% of sample to guarantee the meaningfulness of subgroup and avoid inaccurate estimates of class sizes.

The characteristics of different classes were described by means (standard deviations) for continuous variables and frequencies (percentages) for categorical variables. Subsequently, binary logistic regression models were computed to compare the risk of incident one-year adverse outcomes, including ADL disability, IADL disability, hospitalization and poor QoL, among baseline latent classes of intrinsic capacity, with adjustment for all covariates. Odds ratios (ORs) and 95% confidence intervals (CIs) were reported for each regression model.

Results

Of the 781 participants with available adverse outcomes, 9 missed data on locomotion domain, 2 on vitality domain, 3 on psychological domain, 4 on educational level, and 11 on income, resulting in 756 included for further analyses. More information about analyzed sample has been described in our previously published paper [Blinded for Review]. Between the analyzed sample ($n = 756$) and baseline sample ($n = 1235$), no differences were observed in the baseline characteristics, including socio-demographics characteristics and intrinsic capacity domains, with an exception that the analyzed sample had a higher probability of multimorbidity than the baseline sample [Blinded for Review].

Determining the optimal model

Table 1 shows the fit indices for the one to five-class models. According to model fit criteria, the best-fitting model was three-class one, which yielded the lowest values of aBIC (aBIC = 3460.19) as well as the relatively lower value of BIC (BIC = 3523.70). Although the two-class model yielded the lowest values of BIC, the difference between BIC of two-class model (3520.83) and three-class model

Table 2

Average Latent Class Probabilities for Most Likely Latent Class Membership (Row) by Latent Class (Column)

	Class 1	Class 2	Class 3
Class 1	0.733	0.067	0.2
Class 2	0.087	0.779	0.134
Class 3	0.018	0.071	0.911

Note. Class 1, relatively healthy; Class 2, sharp declines in sensory domain; Class 3, sharp declines in locomotion, psychological, cognition and vitality domains.

(3523.70) was less than 10, indicating that the two-class model did not fit the data better than the three-class one. Moreover, the entropy value of the three-class model reached the peak (Entropy = 0.79); each latent class of the three-class model covered at least 5% of participants (79.4%, 12.8%, and 7.8%) as well as had distinct and substantial interpretability. After the three-class model was established, participants were assigned to the class in which he or she had the highest calculated probability of membership, and the average posterior probabilities ranged from 0.73 to 0.91 across three classes (Table 2), above 0.7 of the criteria of the optimal fitness.

Characteristics of class members

Compared to the other two classes, Class 1 (600/756, 79.4%) had relatively low decline ratios in all intrinsic capacity domains and was therefore labeled as “relatively healthy”; Class 2 (97/756, 12.8%) had particularly high decline rates in sensory domain and moderate decline in locomotion and was therefore named “sharp declines in sensory domain”; Class 3 (59/756, 7.8%) was characterized as showing high decline rates in every domain of intrinsic capacity, especially in locomotion, psychological, cognition and vitality domains, and was therefore named “sharp declines in locomotion, psychological, cognition and vitality domains” (Figure 1).

Characteristics for each class are displayed in Table 3. Older adults in the Class 2 and Class 3 were older, mateless, less educated, as well as had higher probability of multimorbidity and less social support compared with those in Class 1.

Comparison of One-year Adverse Outcome Risks among Classes

Table 4 shows the results of logistic regression analyses comparing the risk of adverse outcomes among three classes, with Class 1 as the reference group that was the largest class representing the majority of the population and showing the overall intrinsic capacity profile. Both Class 2 and Class 3 were at greater risk of ADL (OR = 2.38 - 3.75, 95% CI: 1.32 - 7.49) and IADL disabilities (OR = 2.82 - 2.93, 95% CI: 1.17 - 6.79) than Class 1. Moreover, compared with Class 1, Class 2 was twice as likely to have poor PCS (OR = 1.94, 95% CI: 1.04 - 3.03), and Class 3 had triple greater odds of experiencing poor PCS (OR = 3.01, 95% CI: 1.48, 6.12). Furthermore, older adults in Class 3 were twice as likely to be hospitalized during the one-year follow-up relative to Class 1 (OR = 2.26, 95% CI: 1.23 - 4.16). Other significant differences in risk of incident adverse outcomes were not observed across classes.

Table 1

Fit indices of latent class analysis on intrinsic capacity

Estimated classes	Log (L)	BIC	aBIC	Entropy	Relative sizes of classes
1	-1769.79	3579.35	3560.29	NA	1
2	-1717.34	3520.83	3479.55	0.51	0.243; 0.757
3	-1695.57	3523.70	3460.19	0.79	0.078; 0.794; 0.128
4	-1686.33	3551.62	3465.89	0.77	0.026; 0.143; 0.091; 0.739
5	-1681.56	3588.48	3480.51	0.76	0.616; 0.262; 0.005; 0.093; 0.024

Note. Log (L) = Log likelihood; BIC = Bayesian information criterion; aBIC = adjusted BIC.

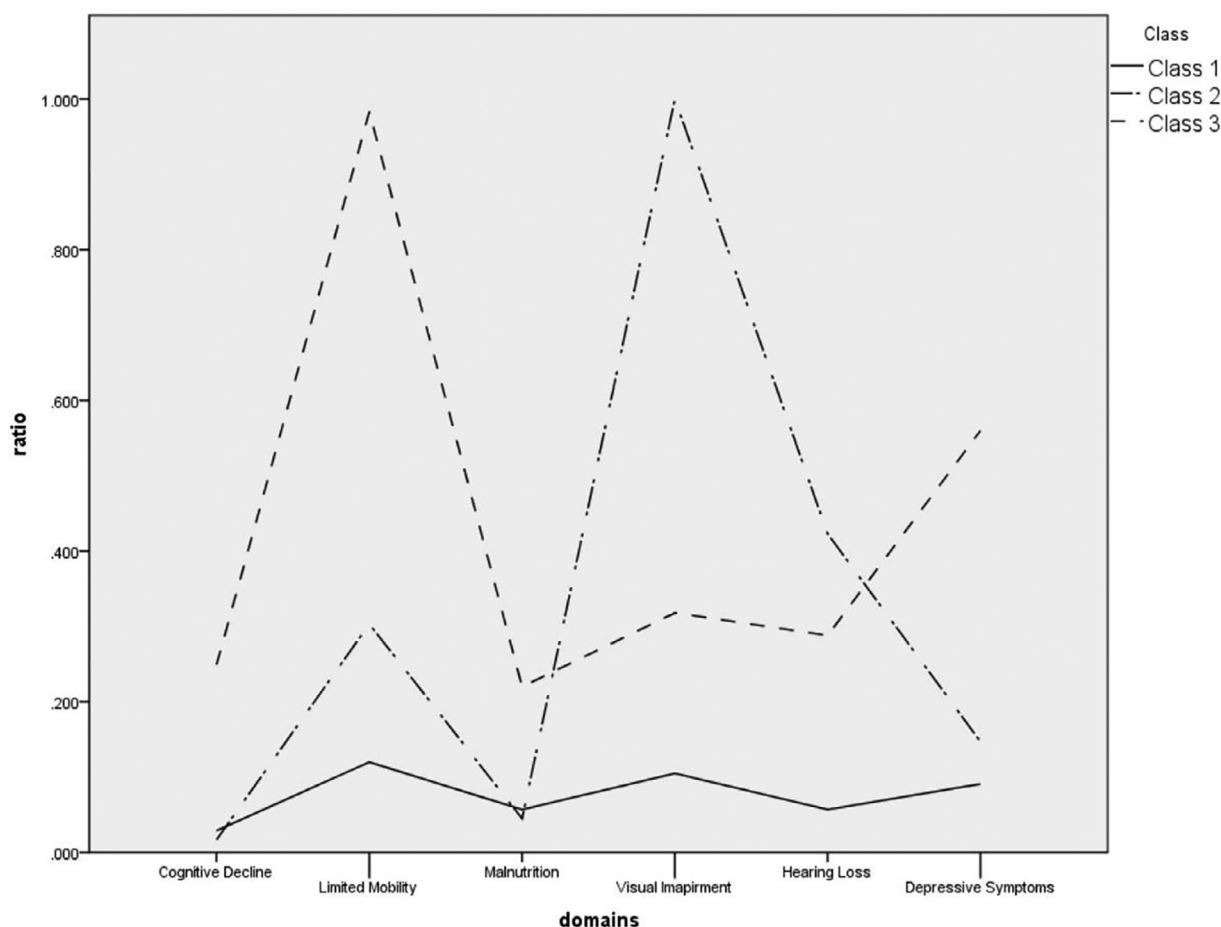


Figure 1. Probability estimates for three class solution

Discussion

This study identified three patterns of declines in intrinsic capacity among community-dwelling older adults, namely “relatively healthy”, “sharp declines in sensory domain”, and “sharp

declines in locomotion, psychological, cognition and vitality domains”. Subgroups with sharp declines in specific combinations of domains had higher risks of one-year adverse outcomes than the relatively healthy subgroup, including disabilities, hospitalization and poor PCS.

Table 3
Baseline characteristics of the total sample and participants assigned to three classes

Characteristics	Total (n = 756)	Class 1 (n = 600)	Class 2 (n = 97)	Class 3 (n = 59)	χ^2 / F	p
Age (y, Mean (SD))	69.3 (6.6)	68.6 (5.9)	71.1 (8.3)	73.1 (8.8)	10.944	< 0.001*
Sex					2.496	0.287
Male	238 (31.5)	197 (32.8)	26 (26.8)	15 (25.4)		
Female	518 (68.5)	403 (67.2)	71 (73.2)	44 (74.6)		
Marital status					9.286	0.010
With a mate	574 (75.9)	470 (78.3)	64 (66.0)	40 (67.8)		
Mateless	182 (24.1)	130 (21.7)	33 (34.0)	19 (32.2)		
Educational level					23.729	< 0.001*
Primary school or lower	240 (31.7)	174 (29.0)	32 (33.0)	34 (57.6)		
Middle school	222 (29.4)	186 (31.0)	22 (22.7)	14 (23.7)		
High school or higher	294 (38.9)	240 (40.0)	43 (44.3)	11 (18.6)		
Monthly income (CNY ^a)					7.068	0.132
Low income	253 (33.5)	188 (31.3)	39 (40.2)	26 (44.1)		
Middle income	299 (39.6)	36 (37.1)	246 (41.0)	17 (28.8)		
High income	204 (27.0)	22 (22.7)	166 (27.7)	16 (27.1)		
Multimorbidity					16.688	< 0.001*
No	362 (47.9)	310 (51.7)	32 (33.0)	20 (33.9)		
Yes	394 (52.1)	290 (48.3)	65 (67.0)	39 (66.1)		
Social support (Mean (SD))	38.2 (7.2)	39.3 (6.8)	34.4 (7.1)	33.5 (7.2)	36.935	< 0.001*

Note. Class 1, relatively healthy; Class 2, sharp declines in sensory domain; Class 3, sharp declines in locomotion, psychological, cognition and vitality domains.

^a CNY, China Yuan, \$US 1 = CNY 6.42. The division of low, middle and high income is based on the tertiles income of the analyzed sample.

* $p < .05$

Table 4Comparison of the risk of adverse outcomes among three classes^a

	Class 2		Class 3	
	OR (95% CI)	p	OR (95% CI)	p
ADL disability	2.38 (1.32–4.28)	0.004*	3.75 (1.88–7.49)	< 0.001*
IADL disability	2.93 (1.38–6.24)	0.005*	2.82 (1.17–6.79)	0.021*
Hospitalization	1.21 (0.73–2.00)	0.467	2.26 (1.23–4.16)	0.009*
Poor PCS	1.94 (1.04–3.63)	0.037*	3.01 (1.48–6.12)	0.002*
Poor MCS	1.29 (0.70–2.37)	0.421	1.18 (0.53–2.62)	0.689

Notes. Class 2, sharp declines in sensory domain; Class 3, sharp declines in locomotion, psychological, cognition and vitality domains; ADL, Activities of Daily Living; IADL, Instrumental Activities of Daily Living; PCS, physical component summary; MCS, mental component summary; OR = odds ratio; CI = confidence interval.

^a Class 1 (relatively healthy) was specified as the reference group. All models were adjusted for age, sex, marital status, educational level, monthly income, multimorbidity, and social support.

* $p < .05$

The LCA indicated that, based on declines in intrinsic capacity domains, older adults could be classified into three meaningful subgroups. Nearly 80% of participants were assigned to Class 1, “relatively healthy”, which illustrates that the majority of participants had nearly intact intrinsic capacity in our survey. Additionally, Class 2 contained 12.8% of participants with sharp declines in sensory domain and moderate decline in locomotion domain, and the smallest class included 7.8% of participants with sharp declines in locomotion, psychological, cognition and vitality domains. The two prominent patterns of intrinsic capacity declines are consistent with the prevalence of disability in the second China National Sample Survey on Disability (CNSDD), which found that physical and sensory disabilities were the two most common types of disability.²⁴ Another research on patterns of function among Spanish older adults similarly identified “the limited sensory functioning class” with poor sensory function and “limited mental and mobility functioning class” with depression, cognitive decline and severe limited mobility.¹²

The Class 2 was characterized by the presence of high decline rates in the sensory domain, including visual and hearing function, and moderate decline rate in the locomotion domain, in line with previous studies that found high risk of limited mobility among older adults with visual impairment or hearing loss.^{25,26} The decline of sensory function may result in poor balance, poor postural stability, and dull perception of danger, lowering their sense of security to the surrounding environment.^{25,27} Thus, they will reduce their activities to avoid accidental injury, which may consequently lead to the substantial decline of motor function and limited mobility. It is worth noting that older adults with sensory impairment had a particularly low decline rate in psychological domain in our study. This may be explained by the fact that older adults regard sensory impairment as a normal manifestation of aging and maintain a less negative attitude towards sensory decline.²⁸ Even so, great importance should be attached to the visual impairment and hearing loss in older adults, reducing the inconvenience brought by sensory impairment and avoiding further psychological disorders caused by limited mobility.

The Class 3 had most impaired intrinsic capacity, especially in locomotion, psychological, cognition and vitality domains. The high decline rates of locomotion and psychological domains are coincident with the fact that limited mobility and depressive symptoms in older adults often co-occur and worsen each other. On the one hand, limited mobility could increase individuals’ depressive symptoms.²⁹ The reason may be that limited mobility leads older adults to depend more on others in ADL, resulting in sense of uselessness and low self-esteem as well as ultimately geriatric depressive symptoms.³⁰ On the other hand, depressed older adults often experience a series of somatic symptoms (e.g., fatigue and pain), which are associated with limited mobility.³¹ Therefore, multidisciplinary interventions that

combine physical exercise and psychological therapy may induce additive or synergistic effects to effectively improve physical and mental health through addressing both biological and psychological components of the modern medical model. The Class 3 also had obvious decline rates in cognition domain relative to Class 1 and Class 2, respectively. Previous research found that older adults with limited mobility or depressive symptoms were more likely to suffer from cognitive decline.^{32,33} The reason why limited mobility is related to cognitive decline may be that declines in locomotion and cognitive domains are driven by a unifying process, namely aging. Motor fitness has been found to be positively associated with higher-level cognitive processes such as attention, visual and spatial information as well as executive functions.³⁴ The links between depressive symptoms and cognitive decline, however, were attributed to multiple biological mechanisms such as vascular diseases, hippocampal atrophy and inflammatory changes.³⁵ Furthermore, malnutrition in older adults has been found to be an important problem related to limited mobility, cognitive impairment and depressive symptoms.^{36–38} This was also supported by our finding that four conditions co-occurred, indicating the need for integrated care of older people in primary care.

Compared with Class 1, the Class 2 and Class 3 had higher risks of disabilities and poor PCS, which may be attributed to the relationship between these three adverse outcomes and the obvious impairment of locomotion in Class 3 as well as sensory domain in Class 2. Consistent with our results, previous studies have found that poor physical performance, such as weak lower extremity strength and slow gait speed, could predict worsening disability³⁹ and poor PCS.⁴⁰ Sensory impairment was also associated with disabilities,^{41,42} PCS and MCS in previous studies.^{43,44} The reason why our study did not find a higher risk of poor MCS among sensory impaired participants might be that Chinese older people are usually more concerned about physical-related quality of life than psychological consequences.

Furthermore, the risk of hospitalization in Class 3 was twice than that in Class 1, but this difference was not significant between Class 2 and Class 1. The possible explanation is that older adults tend to consider visual impairment and hearing loss as natural aging phenomena that cannot be intervened, and are not aware of the existing effective treatment methods of sensory impairment.⁴⁵ Besides, instead of hospitalization, wearing hearing aids is the most widespread treatment of age-related hearing loss.⁴⁶ The high risk of hospitalization among older adults in Class 3 may be mainly attributed to their limited mobility that was an independent predictor of hospitalization.⁴⁷ In terms of the physiological mechanism, since physical performance depends on the functioning of musculoskeletal, nervous and cardiopulmonary systems, limited mobility may be a signal of abnormalities in these systems.⁴⁸ Thus, limited mobility reflects a heavy burden of potential preclinical diseases or physiological abnormalities. Besides, the cognition, vitality and psychological domains of Class 3 were also more damaged relative to the other two classes. Actually, previous studies have identified cognitive decline, malnutrition and depressive symptoms as predictors of hospitalization in later life.^{49–51}

Strengths and Limitations

To our knowledge, this is the first study to apply LCA to identify profiles of intrinsic capacity, and explore the association between latent classes of intrinsic capacity and adverse outcomes. Although the ICOPE is anticipated to develop personalized care plans for older adults on the basis of their needs, values and preferences,¹³ its implementation is indeed limited by medical resources and regional development level. The LCA focuses on the heterogeneity of the population and divides older adults into different subgroups according to their declines in intrinsic capacity domains. Each of the three statistically and clinically distinct classes contained a group of homogeneous

older adults with similar health status and declines in intrinsic capacity, deserving a similar series of targeted interventions and health management. This kind of targeted health intervention at the level of homogeneous population is significant for the early implementation of ICOPE strategy. On the one hand, it contributes to improving the utilization efficiency of healthcare resources and alleviating the shortage of medical human resources, especially suitable for underdeveloped settings with scarce healthcare resources. On the other hand, carrying out simultaneous intervention for homogeneous older adults may achieve peer support, mutual encouragement, supervision and promotion, which is one of the key actions for achieving older-people-centred and integrated care.²

This study has several limitations. As for the LCA itself, although our fit indices and classification accuracy met the established criteria, there was still some degree of uncertainty of latent class membership because the membership probability of each individual in the specific class (0.73 - 0.91) failed to equal one. Thus, the results should be interpreted cautiously. Also, in any LCA, the characteristics of the sample and variables have some influence on the nature and prevalence of latent classes. Further studies are needed to determine whether these patterns of intrinsic capacity in our research could be replicated in other representative samples. Second, our measurement of sensory function is based on dichotomized self-report items rather than objective assessment, where potential measurement bias may limit the accuracy of our results. Nevertheless, the significant correlation between subjective and objective sensory function,⁵⁰ to some extent, may guarantee the credibility of findings. Finally, since our interest was to examine the risk of various one-year adverse outcomes in older adults with different intrinsic capacity patterns at baseline, we did not identify the dynamic patterns of intrinsic capacity longitudinally. Further research can classify the longitudinal trajectories of intrinsic capacity as well as examine its relationship with adverse outcomes.

Implications

Our findings have several implications for healthcare practitioners. First, typologies of intrinsic capacity suggest that it is not meaningful to simply count how many domains decline an individual has. Understanding patterns of intrinsic capacity can help practitioners identify older adults at high risk of adverse outcomes. This study found that nearly 8% of the participants reported high rates of decline in most intrinsic capacity domains, and they were at high risk of various adverse outcomes including disabilities, hospitalization and poor PCS. Although they accounted for a much smaller proportion of the sample, the size of this class was substantial, which suggests that 8-in-100 older adults may have severe vulnerabilities when it comes to their intrinsic capacity. On the other hand, what cannot be ignored is that 12.8% of the participants were divided into “sharp declines in sensory domain”, coinciding with the fact that self-assessed sensory loss are highly prevalent conditions among Chinese older adults.⁵² Second, interventions for the two subgroups of older adults ought to have quite different emphases. Older adults with sharp declines in multiple domains should receive integrated interventions targeting physical exercise rehabilitation, psychological intervention and cognitive training. As for older adults with sharp decline in sensory domain, instead of negatively regarding sensory impairment as a nonfatal and inevitable aspect of aging, healthcare providers as well as older adults themselves should face up to the progressive threat of visual impairment and hearing loss to physical and mental health and take effective strategies. Generally, in view of distinct impaired domains and risk of adverse outcomes by the patterns of intrinsic capacity, the interventions should be tailored for heterogeneous groups.

Conclusions

This study identifies three patterns of intrinsic capacity (i.e., “relatively healthy”, “sharp declines in sensory domain”, and “sharp declines in locomotion, psychological, cognition and vitality domains”), where older adults have differential risks of one-year adverse outcomes, including disabilities, hospitalization and poor QoL. It suggests that older adults can be stratified into heterogeneous groups and deserve different targeted interventions to delay or even prevent the onset of adverse outcomes.

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Conflicts of interest

The authors declare that they have no conflict of interest.

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References

1. Araujo de Carvalho I, Epping-Jordan J, Pot AM, et al. Organizing integrated health-care services to meet older people's needs. *Bull World Health Organ*. Nov 1 2017;95 (11):756–763. <https://doi.org/10.2471/blt.16.187617>.
2. WHO. *World Report on Ageing and Health*. April 1, 2021. Accessed: <https://www.who.int/ageing/events/world-report-2015-launch/en/>.
3. Cesari M, de Carvalho IA, Thiyagarajan JA, et al. Evidence for the domains supporting the construct of intrinsic capacity. *Jf Gerontol Ser a-Biol Sci Med Sci*. Dec 2018;73(12):1653–1660. <https://doi.org/10.1093/gerona/gly011>.
4. Borras C, Ingles M, Mas-Bargues C, et al. Centenarians: An excellent example of resilience for successful ageing. *Mech Ageing Dev*. Mar 2020;186: 111199. <https://doi.org/10.1016/j.mad.2019.111199>.
5. Beard JR, Jotheeswaran AT, Cesari M, de Carvalho IA. The structure and predictive value of intrinsic capacity in a longitudinal study of ageing. *Bmj Open*. Nov 2019;9 (11):e026119. <https://doi.org/10.1136/bmjopen-2018-026119>.
6. Zeng X, Shen S, Xu L, et al. The impact of intrinsic capacity on adverse outcomes in older hospitalized patients: a one-year follow-up study. *Gerontology*. 2021:1–9. <https://doi.org/10.1159/000512794>.
7. Charles A, Buckinx F, Locquet M, et al. Prediction of adverse outcomes in nursing home residents according to intrinsic capacity proposed by the world health organization. *J Gerontol Ser A, Biol Sci Med Sci*. 2020;75(8):1594–1599. <https://doi.org/10.1093/gerona/gz218>. 2020-Jul-13https://doi.org/10.1093/gerona/gz218.
8. Sinha P, Calfee CS, Delucchi KL. Practitioner's guide to latent class analysis: methodological considerations and common pitfalls. *Crit Care Med*. Jan 1 2021;49(1): e63–e79. <https://doi.org/10.1097/ccm.0000000000004710>.
9. Hendryx M, Chojenta C, Byles JE. Obesity Risk Among Young Australian Women: A Prospective Latent Class Analysis. *Obesity (Silver Spring, Md)*. Jan 2020;28(1):154–160. <https://doi.org/10.1002/oby.22646>.
10. Larsen FB, Pedersen MH, Friis K, Glümer C, Lasgaard M. A latent class analysis of multimorbidity and the relationship to socio-demographic factors and health-related quality of life, a national population-based study of 162,283 Danish adults. *PLoS One*. 2017;12(1): e0169426. <https://doi.org/10.1371/journal.pone.0169426>.
11. Olaya B, Moneta MV, Caballero FF, et al. Latent class analysis of multimorbidity patterns and associated outcomes in Spanish older adults: a prospective cohort study. *BMC Geriatr*. Aug 18 2017;17(1):186. <https://doi.org/10.1186/s12877-017-0586-1>.
12. de la Torre-Luque A, Cabello M, Lara E, et al. Functioning profiles in a nationally representative cohort of Spanish older adults: a latent class study. *Health Soc Care Commun*. Nov 2020;28(6):2190–2198. <https://doi.org/10.1111/hsc.13031>.
13. WHO. *Integrated care for older people (ICOPE): guidance for person-centred assessment and pathways in primary care*. April 1, 2021. Accessed: <https://apps.who.int/iris/handle/10665/326843>.
14. Pfeiffer E. A short portable mental status questionnaire for the assessment of organic brain deficit in elderly patients. *J Am Geriatr Soc*. Oct 1975;23(10):433–441. <https://doi.org/10.1111/j.1532-5415.1975.tb00927.x>.
15. Speer DC. Clinically significant change: Jacobson and Truax (1991) revisited. *J Consult Clin Psychol*. Jun 1992;60(3):402–408. <https://doi.org/10.1037/0022-006x.60.3.402>.

16. Kriegsman DM, Deeg DJ, Stalman WA. Comorbidity of somatic chronic diseases and decline in physical functioning; the Longitudinal Aging Study Amsterdam. *J Clin Epidemiol*. Jan 2004;57(1):55–65. [https://doi.org/10.1016/s0895-4356\(03\)00258-0](https://doi.org/10.1016/s0895-4356(03)00258-0).
17. Katz S, Ford AB, Moskowitz RW, Jackson BA, Jaffe MW. Studies of illness in the aged. The index of ADL: a standardized measure of biological and psychosocial function. *JAMA*. Sep 21 1963;185:914–919. <https://doi.org/10.1001/jama.1963.03060120024016>.
18. Lawton MP, Brody EM. Assessment of older people: self-maintaining and instrumental activities of daily living. *Gerontologist*. Autumn. 1969;9(3):179–186.
19. Ware Jr. J, Kosinski M, Keller SD. A 12-Item Short-Form Health Survey: construction of scales and preliminary tests of reliability and validity. *Med Care*. Mar 1996;34(3):220–233. <https://doi.org/10.1097/00005650-199603000-00003>.
20. Xiao S. Theoretical basis and research applications of social support rating scale. *J Clin Psychiatry*. 1994;4:98–100.
21. Chong E, Ho E, Baldevarona-Llego J, et al. Frailty in hospitalized older adults: comparing different frailty measures in predicting short- and long-term patient outcomes. *J Am Med Dir Assoc*. May 2018;19(5):450–457 e3. <https://doi.org/10.1016/j.jamda.2017.10.006>.
22. Nylund KL, Asparoutiov T, Muthen BO. Deciding on the number of classes in latent class analysis and growth mixture modeling: A Monte Carlo simulation study. *Struct Equ Modeling*. 2007;14(4):535–569. <https://doi.org/10.1080/10705510701575396>.
23. Lubke GH, Muthén BOJSEM. Performance of factor mixture models as a function of model size, covariate effects, and class-specific parameters. *Structural Equation Modeling A Multidisciplinary Journal*. 2007;14:26–47.
24. Disability OoCNSSo. *Communique on Major Statistics of the Second China Communique on Major Statistics of the Second China National Sample Survey on Disability*. 2006.
25. Campos J, Ramkhalawansingh R, Pichora-Fuller MK. Hearing, self-motion perception, mobility, and aging. *Hear Res*. Nov 2018;369:42–55. <https://doi.org/10.1016/j.heares.2018.03.025>.
26. Swenor BK, Simonsick EM, Ferrucci L, Newman AB, Rubin S, Wilson V. Visual impairment and incident mobility limitations: the health, aging and body composition study. *J Am Geriatr Soc*. Jan 2015;63(1):46–54. <https://doi.org/10.1111/jgs.13183>.
27. Agmon M, Lavie L, Doumas M. The association between hearing loss, postural control, and mobility in older adults: a systematic review. *J Am Acad Audiol*. Jun 2017;28(6):575–588. <https://doi.org/10.3766/jaaa.16044>.
28. Liu W, Liu Q, Hu W. Investigation of hearing and related diseases in 381 aged people. *J Audiol Speech Pathol*. 2010;18:385–386.
29. Picazzo-Palencia E. Depression and mobility among older adults in Mexico: ENSA-NUT 2012. *Hispanic Health Care Int*. Jun 2016;14(2):94–98. <https://doi.org/10.1177/1540415316650846>.
30. Li N, Pang L, Chen G, Song X, Zhang J, Zheng X. Risk factors for depression in older adults in Beijing. *Can J Psychiatry*. Aug 2011;56(8):466–473. <https://doi.org/10.1177/070674371105600804>.
31. Gardener EA, Huppert FA, Guralnik JM, Melzer D. Middle-aged and mobility-limited: prevalence of disability and symptom attributions in a national survey. *J Gen Intern Med*. Oct 2006;21(10):1091–1096. <https://doi.org/10.1111/j.1525-1497.2006.00564.x>.
32. Tolea MI, Galvin JE. The relationship between mobility dysfunction staging and global cognitive performance. *Alzheimer Dis Assoc Disord*. Jul-Sep 2016;30(3):230–236. <https://doi.org/10.1097/wad.0000000000000136>.
33. Zheng F, Zhong B, Song X, Xie W. Persistent depressive symptoms and cognitive decline in older adults. *Brit J Psychiatry: J Mental Sci*. Nov 2018;213(5):638–644. <https://doi.org/10.1192/bjp.2018.155>.
34. Voelcker-Rehage C, Godde B, Staudinger UM. Physical and motor fitness are both related to cognition in old age. *Eur J Neurosci*. Jan 2010;31(1):167–176. <https://doi.org/10.1111/j.1460-9568.2009.07014.x>.
35. Byers AL, Yaffe K. Depression and risk of developing dementia. *Nat Rev Neurol*. May 3 2011;7(6):323–331. <https://doi.org/10.1038/nrneurol.2011.60>.
36. Wei J, Fan L, Zhang Y, et al. Association between malnutrition and depression among community-dwelling older Chinese adults. *Asia Pac J Public Health*. Mar 2018;30(2):107–117. <https://doi.org/10.1177/1010539518760632>.
37. Yu W, Yu W, Liu X, et al. Associations between malnutrition and cognitive impairment in an elderly Chinese population: an analysis based on a 7-year database. *Psychogeriatrics*. Jan 2021;21(1):80–88. <https://doi.org/10.1111/psyg.12631>.
38. Torres de Araújo JR, Macedo Ferreira LMB, Jerez-Roig J, Costa de Lima K. Mobility limitation in older adults residing in nursing homes in Brazil associated with advanced age and poor nutritional status: an observational study. *J Geriatric Physic Therapy*. 2001. <https://doi.org/10.1519/jpt.0000000000000301>. Mar 12 2021.
39. Wang DXM, Yao J, Zirek Y, Reijnierse EM, Maier AB. Muscle mass, strength, and physical performance predicting activities of daily living: a meta-analysis. *J Cachexia Sarcopenia Muscle*. Feb 2020;11(1):3–25.
40. Trombetti A, Reid KF, Hars M, et al. Age-associated declines in muscle mass, strength, power, and physical performance: impact on fear of falling and quality of life. *Osteoporos Int*. Feb 2016;27(2):463–471.
41. Raina P, Wong M, Massfeller H. The relationship between sensory impairment and functional independence among elderly. *BMC Geriatr*. May 7 2004;4:3. <https://doi.org/10.1186/1471-2318-4-3>.
42. Michikawa T, Nishiwaki Y, Kikuchi Y, et al. Gender-specific associations of vision and hearing impairments with adverse health outcomes in older Japanese: a population-based cohort study. *BMC Geriatr*. Nov 22 2009;9:50. <https://doi.org/10.1186/1471-2318-9-50>.
43. Gao J, Hu H, Yao L. The role of social engagement in the association of self-reported hearing loss and health-related quality of life. *BMC Geriatr*. May 25 2020;20(1):182. <https://doi.org/10.1186/s12877-020-01581-0>.
44. Dev MK, Paudel N, Joshi ND, Shah DN, Subba S. Psycho-social impact of visual impairment on health-related quality of life among nursing home residents. *BMC Health Serv Res*. Aug 15 2014;14:345. <https://doi.org/10.1186/1472-6963-14-345>.
45. Ji F, Chen A, Wang QJH. Balance, Communication. *Hear Loss Aged*. 2015;13:51–57.
46. Perez E, Edmonds BA. A systematic review of studies measuring and reporting hearing aid usage in older adults since 1999: a descriptive summary of measurement tools. *PLoS One*. 2012;7(3):e31831. <https://doi.org/10.1371/journal.pone.0031831>.
47. Callahan KE, Lovato L, Miller ME, et al. Self-reported physical function as a predictor of hospitalization in the lifestyle interventions and independence for elders study. *J Am Geriatr Soc*. Oct 2018;66(10):1927–1933. <https://doi.org/10.1111/jgs.15468>.
48. Ensrud KE, Lui LY, Langsetmo L, et al. Effects of mobility and Multimorbidity on inpatient and postacute health care utilization. *J Gerontol A Biol Sci Med Sci*. Sep 11 2018;73(10):1343–1349. <https://doi.org/10.1093/gerona/glx128>.
49. Abizanda P, Sinclair A, Barcons N, Lizán L, Rodríguez-Mañas L. Costs of malnutrition in institutionalized and community-dwelling older adults: a systematic review. *J Am Med Dir Assoc*. Jan 2016;17(1):17–23. <https://doi.org/10.1016/j.jamda.2015.07.005>.
50. Oosterloo BC, Homans NC, Baatenburg de Jong RJ, Ikram MA, Nagtegaal AP, Goedegeure A. Assessing hearing loss in older adults with a single question and person characteristics: Comparison with pure tone audiometry in the Rotterdam Study. *PLoS One*. 2020;15(1):e0228349.
51. Prina AM, Cosco TD, Denning T, Beekman A, Brayne C, Huisman M. The association between depressive symptoms in the community, non-psychiatric hospital admission and hospital outcomes: a systematic review. *J Psychosom Res*. Jan 2015;78(1):25–33. <https://doi.org/10.1016/j.jpsychores.2014.11.002>.
52. Heine C, Browning CJ, Gong CH. Sensory Loss in China: prevalence, use of aids, and impacts on social participation. *Front Public Health*. 2019;7:5. <https://doi.org/10.3389/fpubh.2019.00005>.
53. Yu J, Si H, Qiao X, et al. Predictive value of intrinsic capacity on adverse outcomes among community-dwelling older adults. *Geriatric Nursing*. 2021;42(6):1257–1263.